

# The ML Project Ladder

A staged path from beginner-clean to research-frontier — each project chosen for the *specific* skill it forces (EDA, relational feature engineering, imbalance, leakage-proof CV, p>>n selection, multimodal & graph deep learning). The final section maps the ladder onto single-cell and spatial transcriptomics, where your own work lives.

★ = directly relevant to single-cell / spatial / cancer-genomics work · Links are platform slugs (prefix with https://) · Compiled for Nishant

**How to climb it.** Don't collect — go **deep on two or three per rung**. Write a real EDA, keep a leakage-proof CV harness, log every experiment, then read the winners' write-ups *after* your own attempt and close the gap. The highest-leverage diagonal for you: **Home Credit** (relational FE) → **Open Problems Multimodal** (single-cell DL) → **HEST-1k / RSNA-MICCAI** (spatial + medical imaging) — which walks you straight into your research frontier.

## 1 · Classification

15  
projects

### Tier 1 — Bridge past Titanic (richer, still clean)

| PROJECT                              | SIZE / SHAPE       | SKILLS IT FORCES                                                | METRIC        | LINK                                                                                                      |
|--------------------------------------|--------------------|-----------------------------------------------------------------|---------------|-----------------------------------------------------------------------------------------------------------|
| <b>Spaceship Titanic</b>             | ~13k rows, 13 cols | Feature extraction (Cabin→deck/side, group IDs), clean CV setup | Accuracy      | <a href="https://kaggle.com/competitions/spaceship-titanic">kaggle.com/competitions/spaceship-titanic</a> |
| <b>Playground Series (any month)</b> | varies (synthetic) | Fast iteration of the full loop: EDA → FE → tune → blend        | AUC / logloss | <a href="https://kaggle.com/competitions/playground-series">kaggle.com/competitions → "Playground"</a>    |

### Tier 2 — Imbalance & anonymized features (FE creativity)

| PROJECT                               | SIZE / SHAPE              | SKILLS IT FORCES                                                                  | METRIC     | LINK                                                                                                                                  |
|---------------------------------------|---------------------------|-----------------------------------------------------------------------------------|------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Credit Card Fraud (ULB)</b>        | 285k rows, 0.17% positive | Extreme imbalance, PR-AUC thinking, resampling, cost-sensitive thresholds         | AUPRC      | <a href="https://kaggle.com/datasets/mlg-ulb/creditcardfraud">kaggle.com/datasets/mlg-ulb/creditcardfraud</a>                         |
| <b>Porto Seguro Safe Driver</b>       | ~595k × 59 (anon)         | Signal engineering on opaque features, imbalance                                  | Norm. Gini | <a href="https://kaggle.com/competitions/porto-seguro-safe-driver-prediction">.../porto-seguro-safe-driver-prediction</a>             |
| <b>Santander Customer Transaction</b> | 200k × 200 (anon)         | Adversarial validation, frequency "magic" features, distrust the obvious          | AUC        | <a href="https://kaggle.com/competitions/santander-customer-transaction-prediction">.../santander-customer-transaction-prediction</a> |
| <b>IEEE-CIS Fraud Detection</b>       | ~590k, 2 tables           | Aggregation features, categorical encoding at scale, memory reduction, time split | AUC        | <a href="https://kaggle.com/competitions/ieee-fraud-detection">kaggle.com/competitions/ieee-fraud-detection</a>                       |

### Tier 3 — Multi-table / industrial scale (the medal tier)

| PROJECT                                  | SIZE / SHAPE                  | SKILLS IT FORCES                                                                             | METRIC             | LINK                                                            |
|------------------------------------------|-------------------------------|----------------------------------------------------------------------------------------------|--------------------|-----------------------------------------------------------------|
| <b>Home Credit Default Risk</b>          | 307k main + 6 related tables  | Relational feature engineering & aggregation — the gold standard                             | AUC                | <a href="#">.../home-credit-default-risk</a>                    |
| <b>American Express Default</b>          | ~5.5M rows time-series, ~50GB | Memory optimization (parquet/int), per-customer time-series aggregation, custom metric       | Amex M             | <a href="#">.../amex-default-prediction</a>                     |
| <b>Microsoft Malware Prediction</b>      | ~8.9M × 80 (mostly cat.)      | Train/test distribution shift, adversarial validation, validation strategy as the whole game | AUC                | <a href="#">.../microsoft-malware-prediction</a>                |
| <b>Otto Group Product Classification</b> | 61k × 93, 9 classes           | Multiclass logloss, stacking & ensembling craft                                              | Multiclass logloss | <a href="#">.../otto-group-product-classification-challenge</a> |

### Tier 4 — High-dimensional bulk genomics ( $p \gg n$ ) ★

| PROJECT                                          | SIZE / SHAPE                | SKILLS IT FORCES                                                                            | METRIC                | LINK                                                         |
|--------------------------------------------------|-----------------------------|---------------------------------------------------------------------------------------------|-----------------------|--------------------------------------------------------------|
| <b>Gene Expression Cancer RNA-Seq (PANCAN) ★</b> | 801 × 20,531, 5 tumor types | $p \gg n$ feature selection, variance/univariate filtering, nested CV vs. selection leakage | Accuracy / F1         | <a href="#">UCI id 401 · fetch_ucirepo(id=401)</a>           |
| <b>CuMiDa — Brain Cancer Expression ★</b>        | small microarray cohorts    | Biologically-informed feature selection, tiny-sample regularization                         | Accuracy              | <a href="#">.../brain-cancer-gene-expression-cumida</a>      |
| <b>Mechanisms of Action (MoA) ★</b>              | ~24k × ~900, 206 labels     | Multi-label classification, label correlation, NN vs. trees                                 | Mean col-wise logloss | <a href="#">kaggle.com/competitions/lish-moa</a>             |
| <b>Golub Leukemia (ALL/AML) ★</b>                | 72 × 7,129, 2 classes       | Why naive CV leaks on $p \gg n$ ; doing it right on a famous tiny set                       | Accuracy              | <a href="#">kaggle.com/datasets/crawford/gene-expression</a> |

## 2 · Regression

9  
projects

### Tier 1 — Consolidate the fundamentals

| PROJECT                                  | SIZE / SHAPE    | SKILLS IT FORCES                                                           | METRIC     | LINK                                                            |
|------------------------------------------|-----------------|----------------------------------------------------------------------------|------------|-----------------------------------------------------------------|
| <b>House Prices: Advanced Regression</b> | 1,460 × 80      | FE, skewed-target log-transform, ordinal encoding, Ridge/Lasso + GBM blend | RMSE (log) | <a href="#">.../house-prices-advanced-regression-techniques</a> |
| <b>Concrete Strength / Bike Sharing</b>  | ~1k / ~17k rows | Residual diagnostics, interaction terms, target transforms                 | RMSE       | <a href="#">UCI · fetch_ucirepo</a>                             |

## Tier 2 — Anonymized, high-dimensional, leakage-prone

| PROJECT                                    | SIZE / SHAPE                 | SKILLS IT FORCES                                                                               | METRIC         | LINK                                                     |
|--------------------------------------------|------------------------------|------------------------------------------------------------------------------------------------|----------------|----------------------------------------------------------|
| <b>Allstate Claims Severity</b>            | 188k × 130                   | Heavy-tailed target transforms, custom objectives, MAE-vs-RMSE strategy, high-cardinality cats | MAE            | <a href="#">.../allstate-claims-severity</a>             |
| <b>Mercedes-Benz Greener Manufacturing</b> | 4,209 × 376 (anon, p≈n)      | Dimensionality reduction as features (PCA/ICA), overfit control on a tiny signal               | R <sup>2</sup> | <a href="#">.../mercedes-benz-greener-manufacturing</a>  |
| <b>Santander Value Prediction</b>          | 4,459 × 4,991 (anon, sparse) | Leakage detection, adversarial validation, skepticism toward "too-good" CV                     | RMSLE          | <a href="#">.../santander-value-prediction-challenge</a> |

## Tier 3 — Time-series & multi-table regression

| PROJECT                                     | SIZE / SHAPE             | SKILLS IT FORCES                                                            | METRIC | LINK                                                       |
|---------------------------------------------|--------------------------|-----------------------------------------------------------------------------|--------|------------------------------------------------------------|
| <b>Rossmann Store Sales</b>                 | ~1M rows                 | Date-derived & external features (holidays/promos) over model complexity    | RMSPE  | <a href="#">.../rossmann-store-sales</a>                   |
| <b>Store Sales — Time Series (Favorita)</b> | ~3M rows, hierarchical   | Lag/rolling features, proper time-based CV, cross-split leakage avoidance   | RMSLE  | <a href="#">.../store-sales-time-series-forecasting</a>    |
| <b>Optiver Realized Volatility</b>          | order-book, high-freq    | Microstructure FE at scale (ties to your swing-trading interest)            | RMSPE  | <a href="#">.../optiver-realized-volatility-prediction</a> |
| <b>Elo Merchant Category</b>                | multi-table transactions | Relational aggregation (like Home Credit) for regression; stubborn outliers | RMSE   | <a href="#">.../elo-merchant-category-recommendation</a>   |

## 3 · Deep Learning

9  
projects

DL shifts the effort: less hand-crafted FE, far more on data pipelines, augmentation, transfer learning, and training stability (LR schedules, mixed precision, early stopping).

### Medical imaging — your turf ★

| PROJECT                                              | SIZE / SHAPE                                | SKILLS IT FORCES                                                                               | METRIC | LINK                                                                    |
|------------------------------------------------------|---------------------------------------------|------------------------------------------------------------------------------------------------|--------|-------------------------------------------------------------------------|
| <b>RSNA-MICCAI Brain Tumor Radiogenomic</b><br>★ GBM | ~2,040 patients, 3D mpMRI (FLAIR/T1/T1c/T2) | Predict MGMT promoter methylation from MRI; 3D CNN, transfer learning, a genuinely hard signal | AUC    | <a href="#">.../rsna-miccai-brain-tumor-radiogenomic-classification</a> |
| <b>Histopathologic Cancer Detection (PCam)</b> ★     | ~220k image patches                         | Entry point to medical-imaging CNNs & transfer learning on tissue                              | AUC    | <a href="#">.../histopathologic-cancer-detection</a>                    |
| <b>SIIM-ISIC Melanoma</b> ★                          | ~33k images + metadata                      | Image+tabular multimodal fusion, heavy imbalance, TTA/augmentation                             | AUC    | <a href="#">.../siim-isic-melanoma-classification</a>                   |

| PROJECT                               | SIZE / SHAPE        | SKILLS IT FORCES                                                  | METRIC           | LINK                                                                                             |
|---------------------------------------|---------------------|-------------------------------------------------------------------|------------------|--------------------------------------------------------------------------------------------------|
| <b>APTOS 2019 Blindness Detection</b> | ~3.6k retina images | Ordinal targets (regression vs. classification), QWK optimization | Quad. weighted k | <a href="https://kaggle.com/aptos2019-blindness-detection">.../aptos2019-blindness-detection</a> |

## Vision, NLP & time-series foundations

| PROJECT                          | SIZE / SHAPE               | SKILLS IT FORCES                                                         | METRIC       | LINK                                                                                                                             |
|----------------------------------|----------------------------|--------------------------------------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------|
| <b>CIFAR-10 / CIFAR-100</b>      | 60k images                 | Build a ResNet from scratch; augmentation, schedulers, regularization    | Accuracy     | <a href="#">torchvision / keras built-in</a>                                                                                     |
| <b>Cassava Leaf Disease</b>      | ~21k images                | Noisy labels, label smoothing, mixup/cutmix, robust training             | Accuracy     | <a href="https://kaggle.com/cassava-leaf-disease-classification">.../cassava-leaf-disease-classification</a>                     |
| <b>Jigsaw Toxic Comment</b>      | ~160k texts, multi-label   | Transformer fine-tuning (BERT/ RoBERTa), per-label threshold calibration | Mean col AUC | <a href="https://kaggle.com/jigsaw-toxic-comment-classification-challenge">.../jigsaw-toxic-comment-classification-challenge</a> |
| <b>CommonLit Readability</b>     | ~2.8k texts                | Transformer regression head, robust CV on small text                     | RMSE         | <a href="https://kaggle.com/commonlitreadabilityprize">.../commonlitreadabilityprize</a>                                         |
| <b>M5 Forecasting — Accuracy</b> | hierarchical Walmart sales | GBM vs. deep sequence models (LSTM/N-BEATS/TFT) on the same problem      | WRMSSE       | <a href="https://kaggle.com/m5-forecasting-accuracy">.../m5-forecasting-accuracy</a>                                             |

## 4 · Single-Cell & Spatial Transcriptomics ★

14  
projects

Your research frontier. These flip the challenge from rows-of-records to **cell-level resolution, spatial graphs, and multimodality**. Tool tie-ins noted: scVI/scANVI, Geneformer/scGPT, STAGATE, GraphST, Squidpy, Cell2location.

### Single-cell — cell-type & disease classification

| PROJECT                                          | SIZE / SHAPE                | SKILLS IT FORCES                                                                         | METRIC              | LINK                                                                   |
|--------------------------------------------------|-----------------------------|------------------------------------------------------------------------------------------|---------------------|------------------------------------------------------------------------|
| <b>Tabula Sapiens / Tabula Muris</b> ★           | 500k+ cells, many tissues   | Cross-tissue cell-type classification; marker-aware features; HVG selection              | Macro-F1            | <a href="#">tabula-sapiens (CZ Biohub)</a> · <a href="#">CELLxGENE</a> |
| <b>Zheng68K (PBMC 68k)</b> ★                     | 68k cells, ~11 immune types | Hard annotation benchmark — highly correlated immune subtypes; the classic stress test   | Accuracy / macro-F1 | <a href="#">10x datasets · scanpy ecosystem</a>                        |
| <b>CELLxGENE Census</b> ★ <b>you submit here</b> | millions of annotated cells | Cross-dataset generalization & batch robustness; query-driven train/test splits          | Accuracy            | <a href="#">cellxgene-census (CZI)</a>                                 |
| <b>COVID-19 / disease PBMC atlas</b> ★           | varies (100k–1M cells)      | Disease-state classification at cell & pseudobulk level; donor-level CV to avoid leakage | AUC                 | <a href="https://covid19cellatlas.org">covid19cellatlas.org</a>        |

## Single-cell — deep learning (multimodal, foundation models)

| PROJECT                                                          | SIZE / SHAPE                      | SKILLS IT FORCES                                                                                                 | METRIC            | LINK                                                        |
|------------------------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------------------------------------------|-------------------|-------------------------------------------------------------|
| <b>Open Problems — Multimodal Integration (NeurIPS '22) ★</b>    | ~300k cells, CITE-seq + Multiome  | Cross-modality prediction (RNA→protein, ATAC→RNA); generalize across donors & later time points                  | Pearson corr.     | <a href="#">.../open-problems-multimodal</a>                |
| <b>Open Problems — Single-Cell Perturbations (NeurIPS '23) ★</b> | PBMC under chemical perturbations | Predict differential expression; generalize to <i>unseen</i> perturbations & cell types; representation learning | Mean rowwise RMSE | <a href="#">.../open-problems-single-cell-perturbations</a> |
| <b>scVI / scANVI (PBMC, cortex) ★</b>                            | standard scRNA datasets           | VAE for dimensionality reduction & batch correction; scANVI = semi-supervised label transfer                     | latent / kNN-acc  | <a href="#">scvi-tools.org</a>                              |
| <b>Geneformer / scGPT fine-tuning ★ frontier</b>                 | Genecorpus ~30M cells (pretrain)  | Fine-tune a single-cell foundation model for cell-type / disease classification; transfer learning               | Accuracy          | <a href="#">huggingface.co/ctheodoris/Geneformer</a>        |

## Spatial — domain detection & spatial classification

| PROJECT                                                     | SIZE / SHAPE                      | SKILLS IT FORCES                                                                                                                       | METRIC   | LINK                                                              |
|-------------------------------------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|----------|-------------------------------------------------------------------|
| <b>DLPFC — spatialLIBD (Maynard 2021) ★ STAGATE/GraphST</b> | 12 Visium sections, manual layers | Spatial domain identification — <i>the benchmark</i> your graph tools (STAGATE, GraphST) are scored on; build/compare the GNN yourself | ARI      | <a href="#">spatial.libd.org/spatialLIBD</a>                      |
| <b>Xenium / MERFISH public data ★</b>                       | 100s–1000s genes, subcellular     | Spatial cell-type classification & niche analysis; integrate with Squidpy neighborhood graphs                                          | Accuracy | <a href="#">10xgenomics.com/datasets</a> · <a href="#">Vizgen</a> |

## Spatial — deep learning (histology ↔ expression, segmentation)

| PROJECT                                               | SIZE / SHAPE                             | SKILLS IT FORCES                                                                                                  | METRIC             | LINK                                                    |
|-------------------------------------------------------|------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------|---------------------------------------------------------|
| <b>HEST-1k ★</b>                                      | 1,200+ ST + H&E WSIs, ~2TB (subsettable) | Predict gene expression from histology; benchmark pathology foundation models; multimodal representation learning | Pearson / Spearman | <a href="#">huggingface.co/datasets/MahmoodLab/hest</a> |
| <b>HuBMAP — Hacking the Kidney (2021) ★</b>           | PAS-stained kidney WSIs                  | FTU (glomeruli) semantic segmentation on giant WSIs; tiling, U-Net/transformer decoders                           | Dice               | <a href="#">kaggle.com/c/hubmap-kidney-segmentation</a> |
| <b>HuBMAP + HPA — Hacking the Human Body (2022) ★</b> | multi-organ WSIs                         | Multi-organ FTU segmentation; domain generalization across tissue types/stains                                    | Dice               | <a href="#">.../hubmap-organ-segmentation</a>           |

| PROJECT                                                | SIZE / SHAPE                    | SKILLS IT FORCES                                                                      | METRIC | LINK                                                     |
|--------------------------------------------------------|---------------------------------|---------------------------------------------------------------------------------------|--------|----------------------------------------------------------|
| <b>HuBMAP — Hacking the Human Vasculature (2023) ★</b> | kidney WSI tiles, sparse annot. | Microvasculature <i>instance</i> segmentation; semi-supervised use of unlabeled tiles | mAP    | <a href="#">.../hubmap-hacking-the-human-vasculature</a> |

## 5 · Working each project like the top 1%

The dataset is the easy part. What separates a strong portfolio from a great one is method discipline. Apply this loop to every rung above.

- 1. Frame & baseline first.** Write down the metric, the leakage risks, and a dumb baseline (majority class / mean) before any modeling. If you can't beat the baseline you have a bug, not a model.
- 2. Lock a leakage-proof CV harness early.** Choose the split that mirrors the test distribution (time-based, group/donor-based for single-cell) and never touch it again. For  $p \gg n$  genomics, feature selection goes *inside* the CV fold.
- 3. EDA with hypotheses, not just plots.** Each chart should answer a question ("is the target shifting over time?", "which features differ train vs. test?"). Adversarial validation is part of EDA.
- 4. Feature engineering before model zoo.** On tabular/relational data, FE out-earns tuning. Log every feature's CV delta.
- 5. Tune deliberately.** Coarse random search → narrow Bayesian/Optuna. Tune the few parameters that matter (for GBMs: `num_leaves`, `learning_rate`, `n_estimators`, regularization).
- 6. Experiment log = your moat.** One row per run: hash, CV, LB, what changed. This is the artifact that makes you reproducible and promotable.
- 7. Read winners last.** Attempt fully, then study the gold-medal write-up and re-implement the one idea you missed. That gap-closing is where the learning compounds.

```
project/
├─ data/           # raw (read-only), interim, processed
├─ notebooks/     # 01_eda 02_features 03_baseline 04_tuning 05_ensemble
├─ src/
│  ├─ data.py      # load + leakage-proof split (group/time aware)
│  ├─ features.py  # all FE; each fn returns df + logs CV delta
│  ├─ cv.py        # single source of truth for folds
│  ├─ train.py     # model + Optuna; writes to experiments.csv
│  └─ metrics.py   # the competition metric, exactly
├─ experiments.csv # run_id, cv_score, lb_score, notes, timestamp
└─ README.md      # problem framing, baseline, current best, next ideas
```

Reuse this skeleton across every project — the muscle memory of a clean harness is itself a top-1% skill. Keep raw data read-only; never let a notebook mutate it.